



VLSI Testing

積體電路測試

Fault Modeling

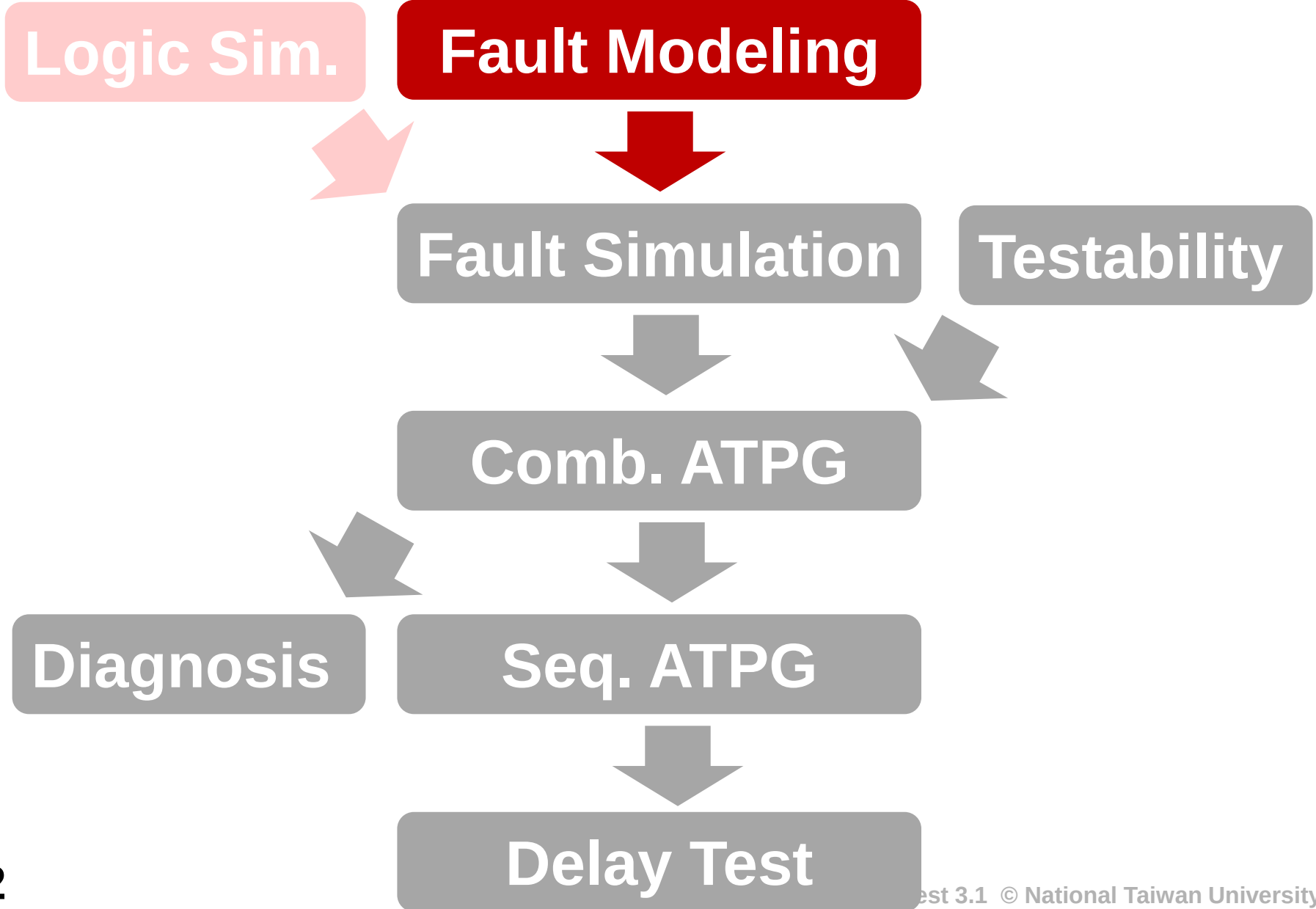
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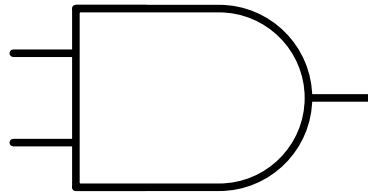
National Taiwan University

Course Roadmap (EDA Topics)



Motivating Problem

- Totally we have 4 possible test patterns: 00, 01, 10, 11
- But your manager says we have time only for 2 test patterns
- Please select **2 best test patterns**. Which? Why?



Why Am I Learning This?

- Fault models **quantify** test quality
- Fault models make test **automation** possible

*“All models are wrong.
But some are useful.”*

(George Box)

Fault Modeling

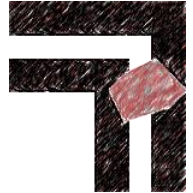
- Introduction
- Fault Models
- Fault Detection
- Fault Coverage
- Conclusion



Definitions

- **Defect**

- ♦ Unintended physical difference between hardware implementation and its intended design
- ♦ Example: unwanted wire (short to ground)

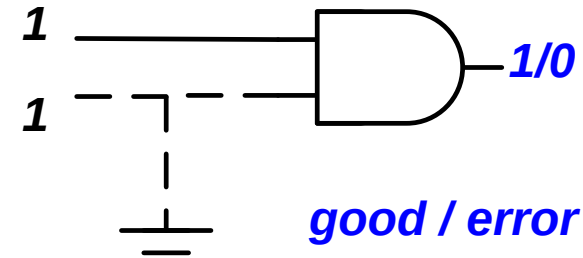


- **Fault**

- ♦ Representation of defects at abstracted logic level
- ♦ Example: b stuck at zero fault

- **Error**

- ♦ Wrong output signal value
- ♦ Example: output = 0, when a=b=1



- **Failure**

- ♦ Deviation from expected behavior
- ♦ Example: computer crash

Defect → Fault → Error → Failure

Examples of Defects

- **Material**
 - ♦ Bulk defect (cracks, crystal imperfections)
 - ♦ Surface impurities
- **Wafer process**
 - ♦ Dust particle → Opens , shorts
 - ♦ Gate oxide pinhole → gate oxide shorts
 - ♦ Bad ion implantation → transistor V_t shift
 - ♦ Mask problem → Missing/extra wires, missing contacts/vias
 - ♦ Lithography problem → Missing/extra wires, missing contacts/vias
 - ♦ Poor etching → ill formed contacts, vias
 - ♦ Careless handling → Oxide breakdown → gate oxide shorts
- **Package problem**
 - ♦ Careless handling → Oxide breakdown
 - ♦ Bad soldering → Contact degradation
 - ♦ Seal leaks
- **Aging (wear-out)**
 - ♦ Overstress → Oxide breakdown
 - ♦ Electromigration → open, short

**There are Many
More Defects ...**

Photos of Defects

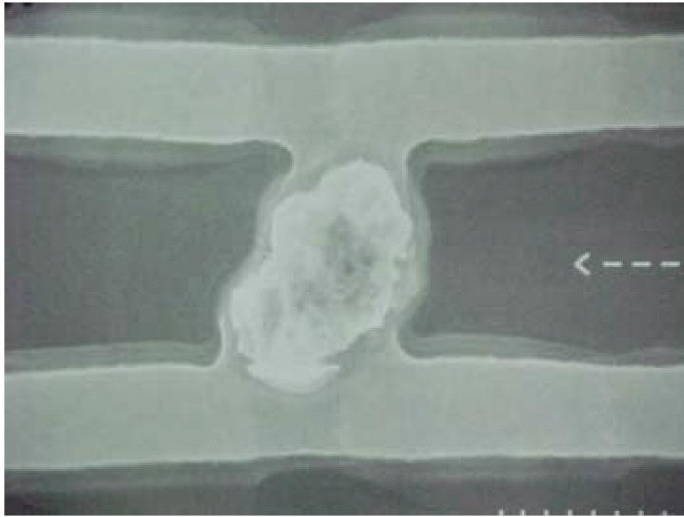
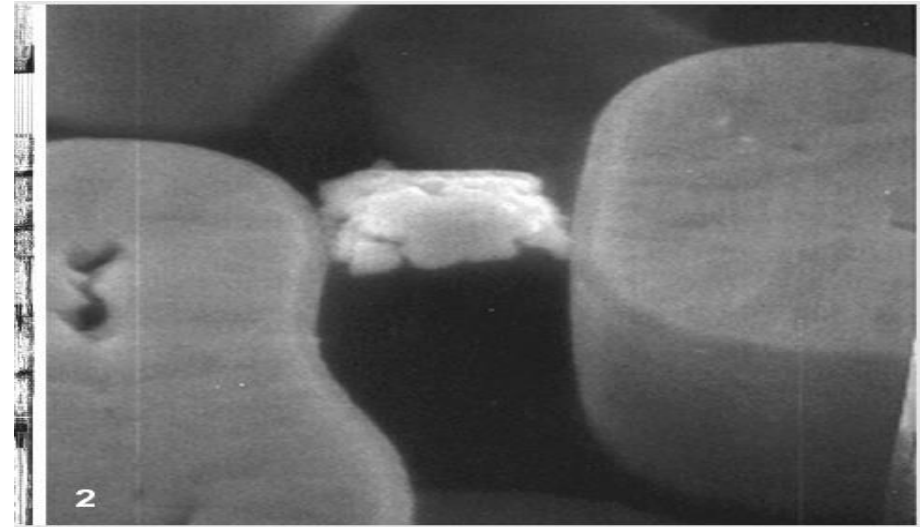
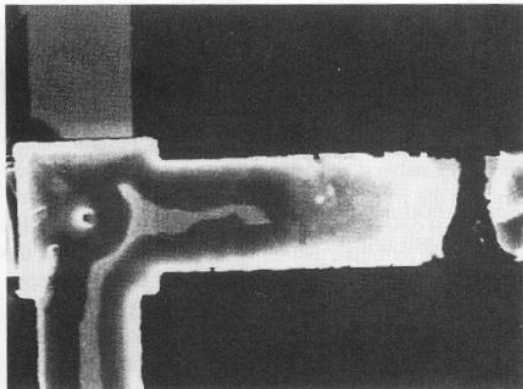


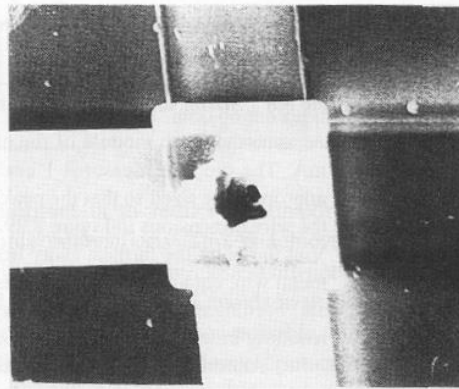
Photo 2-4 Dust-induced Wiring Short



[Vallet IBM 1997]



(a) Line-open failure.



(b) Open failure in contact plug.

Figure 8.30 Electromigration-related failure modes (Courtesy of N. Cheung and A. Tao, U.C. Berkeley).

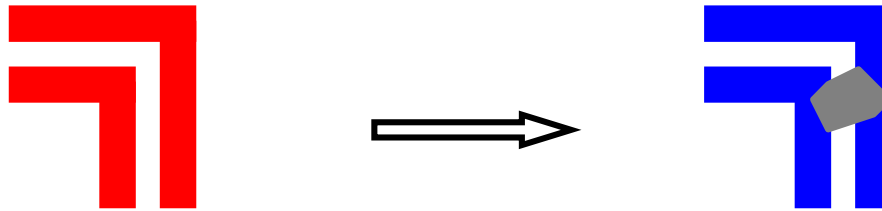


[Nigh IBM 1998]

Random Defects, Systematic Defects

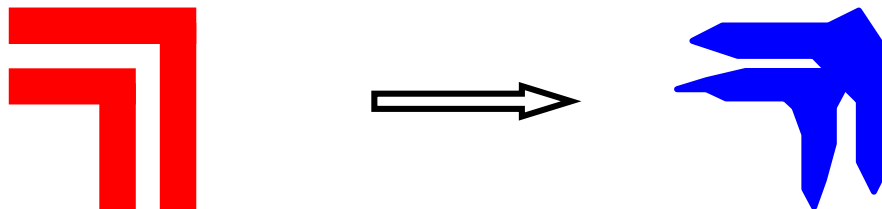
- **Random defects**

- ♦ Caused by random factors such as particles, scratches, ...
- ♦ No correlation across wafers, dies



- **Systematic defects**

- ♦ Caused by deterministic factors such as mask, lithography, ...
- ♦ Strong correlation across wafers, dies



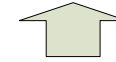
What is Fault Modeling?

- Representing defects at abstracted logic level

Higher level

Behavior

Failure



Error

I/O Signal



Logic

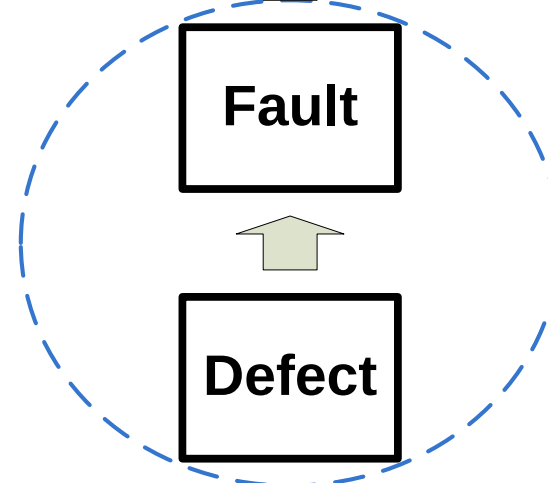
Fault



Physical &
Circuit

Defect

Lower level



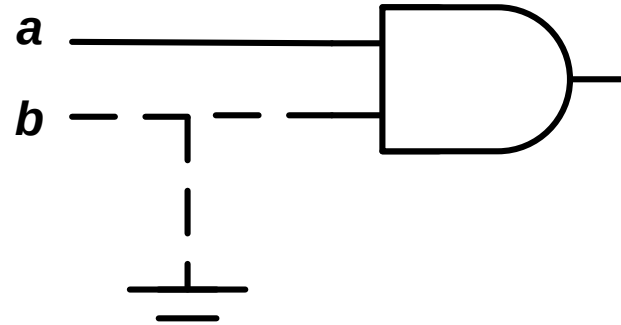
Why Fault Modeling?

- 1. Fault model **quantify** test quality
 - ◆ Defects are hard to handle
 - * How many possible defects in a circuit ? **Way too many**
 - ◆ Number of faults can be easily calculated in a circuit
- 2. Fault models makes test **automation** possible
 - ◆ Automatic test pattern generation (ATPG)
 - * generate test patterns
 - ◆ Fault simulation
 - * evaluate test quality
 - ◆ Automatic diagnosis
 - * locate defects

Automatic Tools Need Fault Model

Test Patterns and Test Sets

- **Test patterns**, also known as (aka): **Test Vectors**
 - ♦ Input Boolean values for specific fault
 - * Expected output often included (but not required)
 - ♦ Example: **b stuck-at zero fault**
 - * Test pattern **a=1, b=1**
- **Test Set** = A set of test patterns
 - ♦ Example: $ab = \{11, 00\}$
- **Test Length** = Number of test patterns in a test set
 - ♦ Example: $ab = \{11, 00\}$, $TL=2$



Shorter TL = Lower Test Cost

Fault Coverage (FC)

$$\text{Fault Coverage} = \frac{\text{number of detected faults}}{\text{number of total faults}} \times 100\%$$

- Between 0-100%
- FC is most widely used quantitative measure of test set quality
 - * Higher fault coverage implies more effective tests
 - 0-70% → not good
 - 70-95% → sometimes acceptable
 - 95-100% → good
- (Standard depends on company and products)
- In Brown & Williams model,
 - ♦ **Defect Level** = $1 - Y^{(1-FC)}$

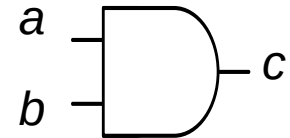
Higher FC = Higher Quality

Quiz

- (Cont'd from P.3) Manager asked you to pick 2 patterns...
- Suppose you picked {11, 00} for AND gate
 - ♦ both output 1 and 0 functions are tested

Q: What is single stuck-at fault coverage?

A:



- Table: total six single stuck-at faults

Input	Fault-free Output	Faulty Output Value with SSF					
<i>a b</i>		<i>a/0</i>	<i>a/1</i>	<i>b/0</i>	<i>b/1</i>	<i>c/0</i>	<i>c/1</i>
0 0	0	0	0	0	0	0	<u>1</u>
1 1	1	<u>0</u>	1	<u>0</u>	1	<u>0</u>	1

*erroneous output values highlighted

Testing With or W/O Fault Models

Comparison	Functional testing test ckt functionality w/o fault model	Structural testing test ckt structure with fault model
Test pattern generation	☹ manual	😊 automatic
Fault coverage	☹ low	😊 high
Test speed	😊 at-speed testing test at specified circuit speed helps to defect delay faults	☹ slow speed testing exercise ckt in different ways from functional mode
Test power	😊 low power	☹ high power
Verification / silicon debug	😊 helps to debug	☹ does not help

Func. and Structural Tests Both Needed

Summary

- Fault modeling
 - ◆ Fault models helps to **quantify** and **automate** testing
 - ◆ **Defect** → **fault** → error → failure
 - ◆ **Test length** = number of test patterns
 - ◆ **Fault coverage** = detected faults / total faults
 - ◆ **Structural test** and **functional test** both needed

